F. Multimodal Inference and generation

F.1 Introduction to Multimodal Representation & Alignment

Part 1: introduction to multimodal AI

Part2: Establish correspondence between two modalities

F.2 Multimodal Fusion

F. 3 Cross-Modal learning


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Part 1 Introduction to Multimodal AI: One object, multiple measurement modalities



Measuring modality **A**

$X^A = \{ \triangle, \triangle, \triangle, \ldots \}$

Measuring modality **B**

$X^B = \{ \bigcirc, \bigcirc, \bigcirc, \ldots \}$

Heterogeneous

Connected

Interacting

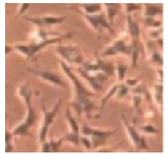
Examples:

scRNAseq





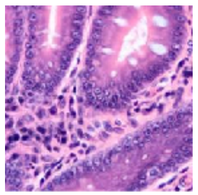
Morphology



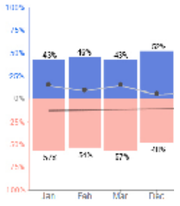


patient

Histology



Clinical record





scRNAseq



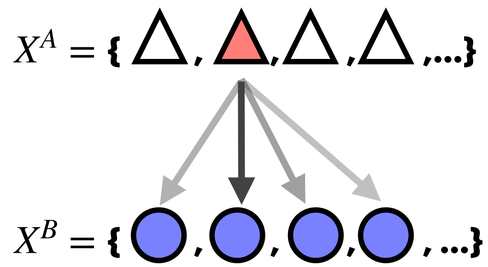
Proteomics



Part 1, introduction: Motivation, what can we do with multiple modalities?

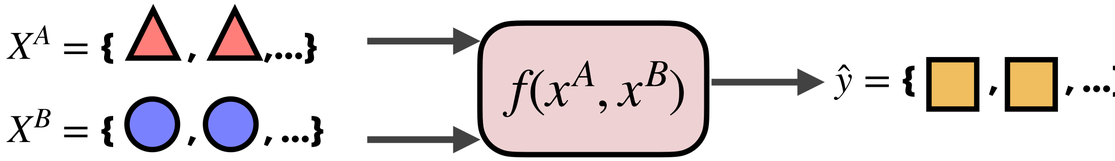
1. Match information: Use one modality to retrieve the corresponding observation in another:

$$x^A \rightarrow \text{matching } x^B$$



2. Combine information: Use both measurements to make a better prediction:

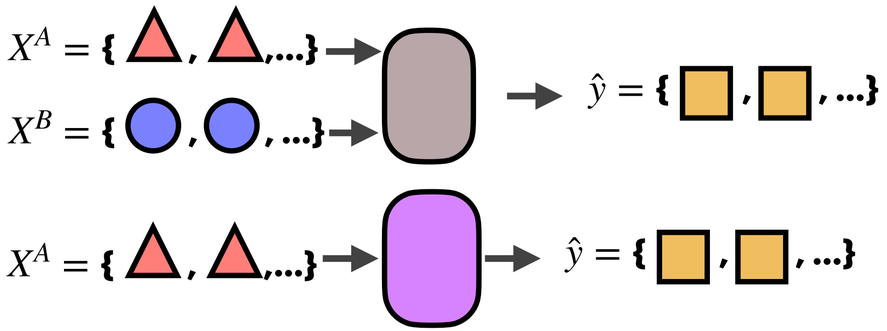
$$\hat{y} = f(x^A, x^B)$$



3. Learn from a modality that will later be missing: Train with both modalities, but deploy with only one:

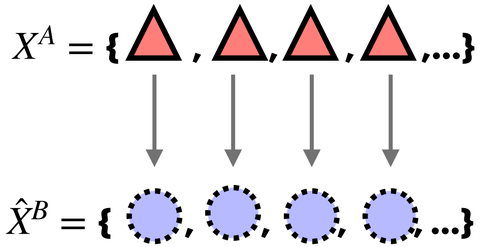
(x^A, x^B) during training

x^A at deployment



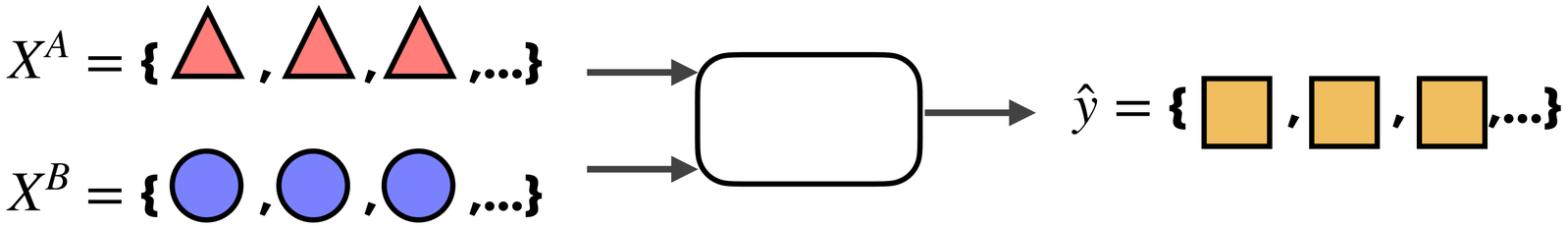
4. Generate or translate information: Use one modality to predict what another modality could look:

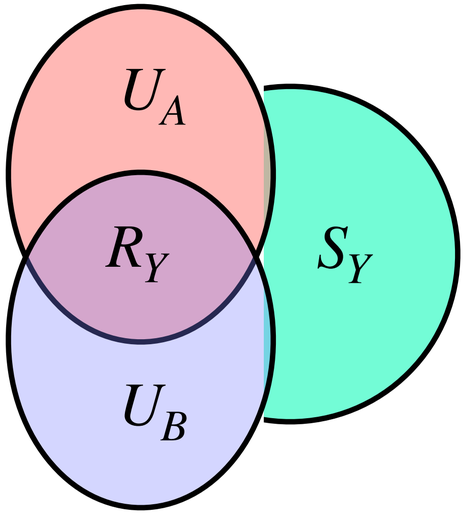
$$\hat{x}^B \sim p_\theta(x^B \mid x^A)$$



Part 1, introduction: What does the each modality add?

How is information about a target Y distributed across X^A and X^B ?





U_A: Unique information available from modality A

U_B: Unique information available from modality B

R_Y: Redundant information: both modalities tell us this about Y

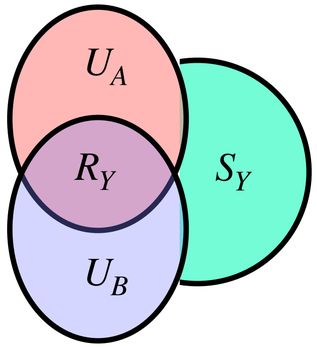
S_Y: Synergistic information: revealed on Y only when A and B are combined

A second modality may be redundant, add unique information,
or reveal a joint pattern that neither modality contains alone:

$$I(Y; X^A, X^B) \approx R_Y + U_A + U_B + S_Y$$

Part 1, introduction: A compass for the next two lecture

R_Y



U_A U_B S_Y

This Lecture

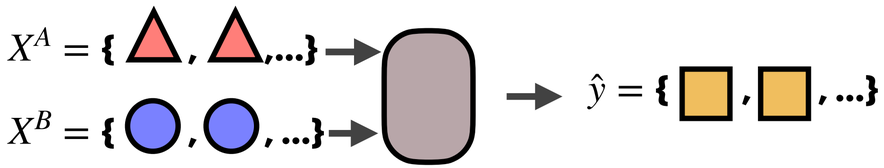
Part 1: introduction

Part2: Establish correspondence

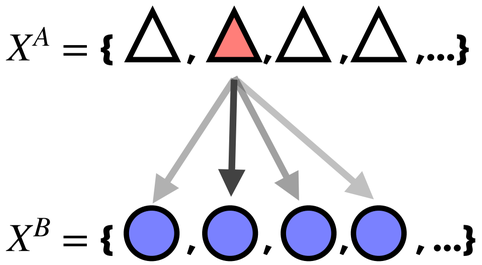
$$x_i^A \longleftrightarrow x_j^B$$

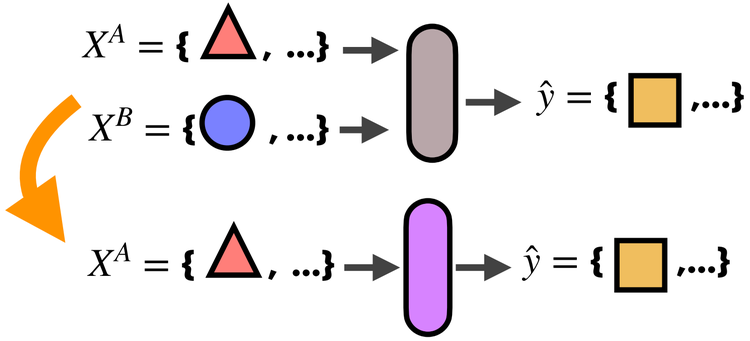
Next Lecture

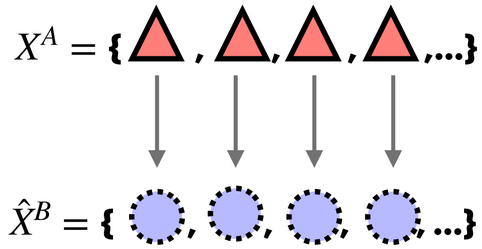
Part A: Use what each modality adds



Part B: What survives when B disappears?







Part2, establish correspondence: Different modalities can live in different data spaces

x_i^A

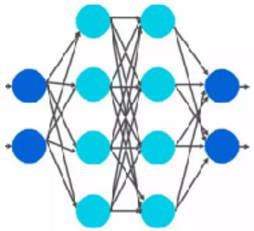
$X^A = \{ \blacktriangle, \blacktriangle, \blacktriangle, ... \}$

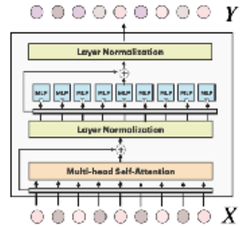
$X^B = \{ \bigcirc, \bigcirc, \bigcirc, ... \}$

x_i^B

Fixed feature vector → **MLP, Transformer**

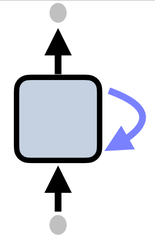
$x_i \in \mathbb{R}^d$





Ordered sequence → **RNN, Transformer**

$x_i = \left(x_i^{(0)}, x_i^{(1)}, \ldots, x_i^{(T_i-1)} \right)$



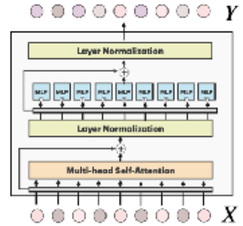
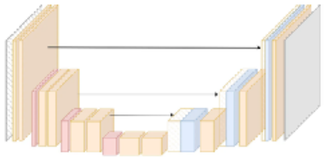
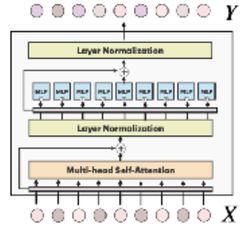


Image / spatial grid → **CNN, Vision Transformer**

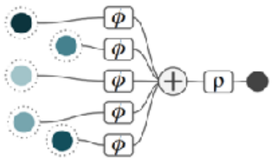
$x_i \in \mathbb{R}^{H_i \times W_i \times C}$

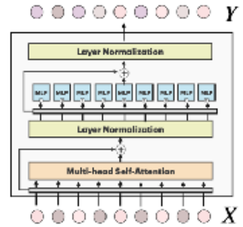




Unordered set → **Deep Sets, Set Transformer**

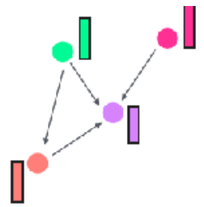
$x_i = \left\{ x_i^{(1)}, x_i^{(2)}, \ldots, x_i^{(n_i)} \right\}$

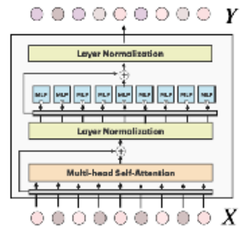




Graph → **GNN, Transformer**

$x_i = G_i = (V_i, E_i, \{x_{iv}\}_{v \in V_i})$

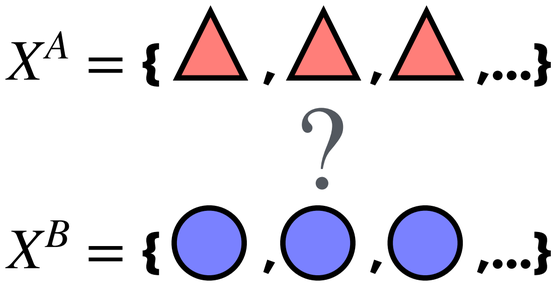




Part2, establish correspondence: Clarifying the setup

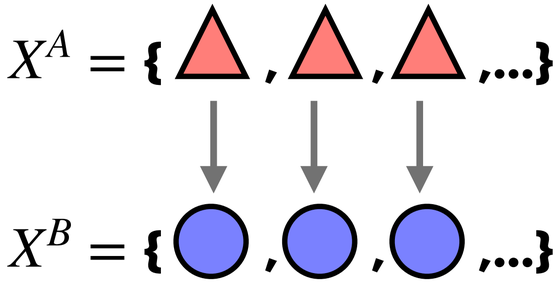
Do we already know which element of X^A (x_i^A) is paired with which element of X^B (x_j^B)?

No, we have unpaired observations:



We need to first infer the correspondence via anchors, distribution matching ,etc.

Yes, correspondence is known, we have **paired observations**:

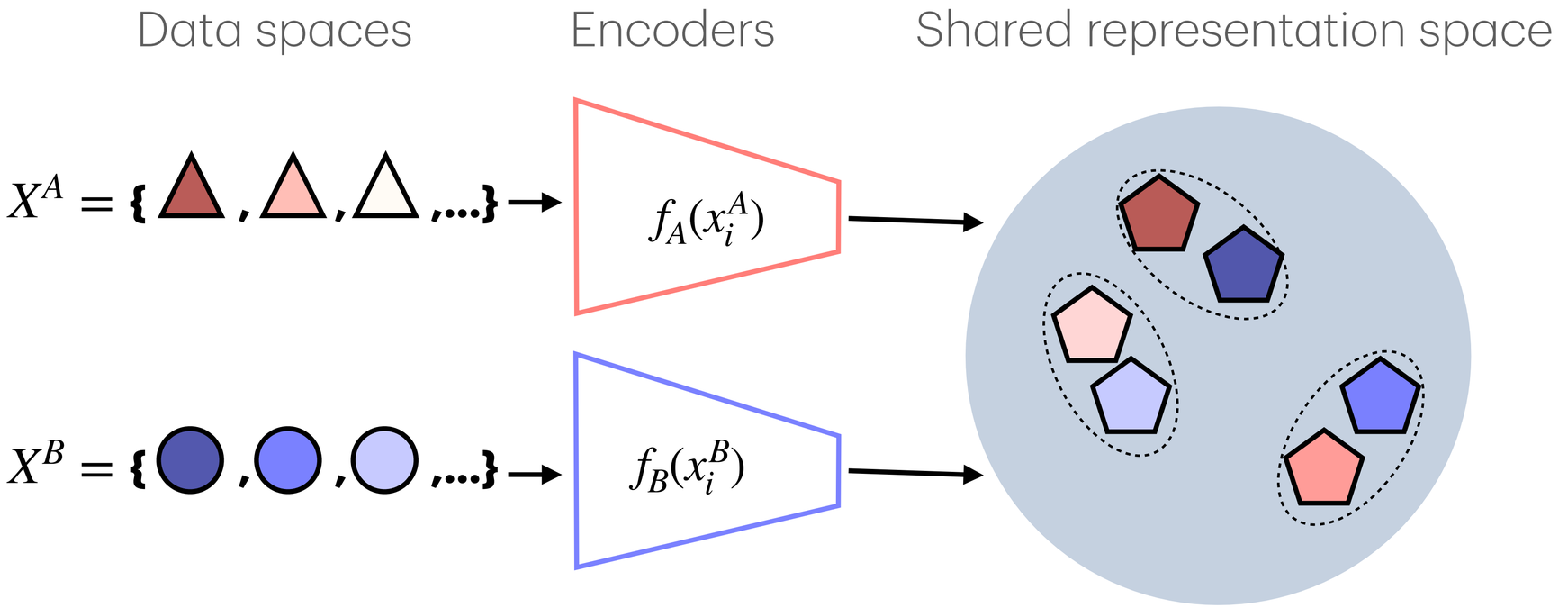


We can use contrastive learning / CLIP to learn a representation consistent with the known match

Rozony, Farhana Zaman, et al. "A systematic review of big data integration challenges and solutions for heterogeneous data sources." *Academic Journal on Business Administration, Innovation & Sustainability* 4.04 (2024): 1-18.

Part2, establish correspondence: The goal of paired alignment

Goal: Map both modalities into a comparable representation space where matched pairs are more similar than mismatched pairs.



Before how, lets see **Why?**

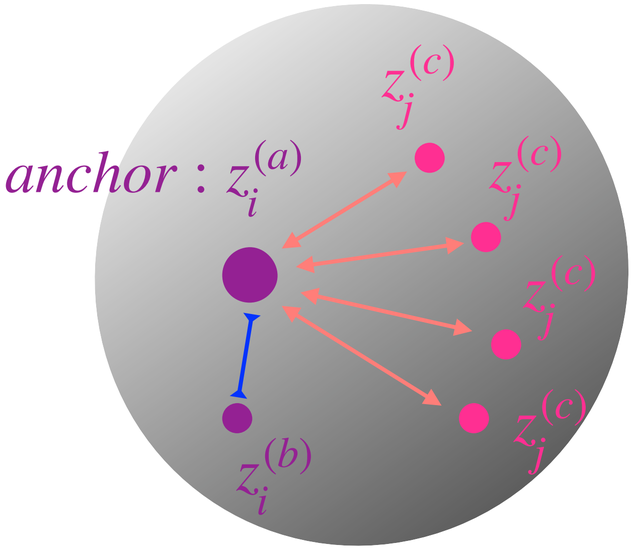
Retrieve matches: Given (x_i^A), find its matching (x_j^B).

Transfer labels: Annotate one modality and transfer labels across modalities.

Deploy with one modality: Learn from paired data, then operate when one modality is unavailable.

Part2, establish correspondence: How do known pairs teach shared representation space?

Reminder: Contrastive learning could be helpful



In contrastive leanring we had:

a) For each anchor point: ($z_i^{(a)}$) we want ($z_i^{(b)}, b \neq a$) to be close,

b) We eant many negative examples ($z_j^{(c)}, j \neq i$) to be far.

How to achieve it:

Define similarity between two points as: $sim(u, v) = \frac{u^\top v}{\|u\|\|v\|}$.

we define loss as:

$$\mathscr{L}_{cont} = -\log \frac{e^{sim\left(z_i^{(a)}, z_i^{(b)}\right)/\tau}}{e^{sim\left(z_i^{(a)}, z_i^{(b)}\right)/\tau} + \sum_{j,c} e^{sim\left(z_i^{(a)}, z_j^{(c)}\right)/\tau}}$$

If Positive points are close to anchors (high similarity), loss decreases

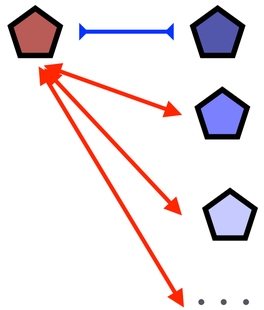
If Negative points are close to anchor (high similarity), loss increases

Part2, establish correspondence: The same contrastive recipe, now across modalities

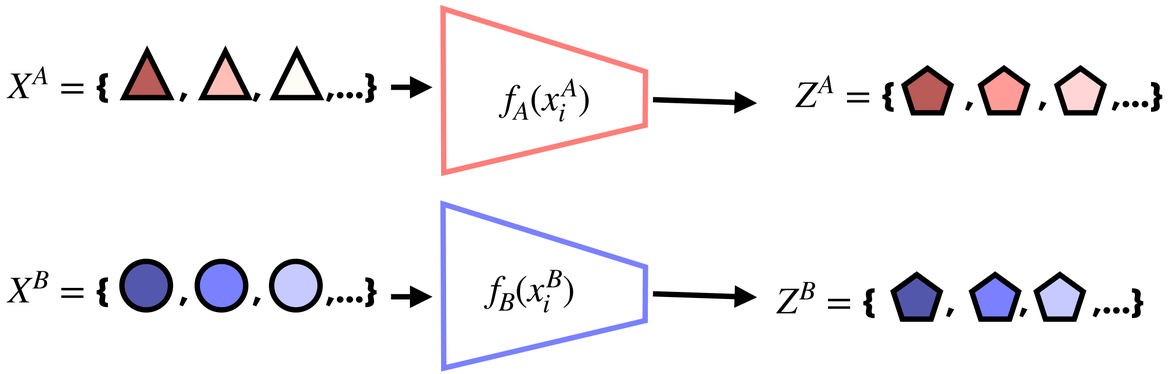
Mapping to shared space

$$z_i^A = f_A(x_i^A),$$
$$z_i^B = f_B(x_i^B)$$

Contrastive:
known pair pull together
other candidates push apart



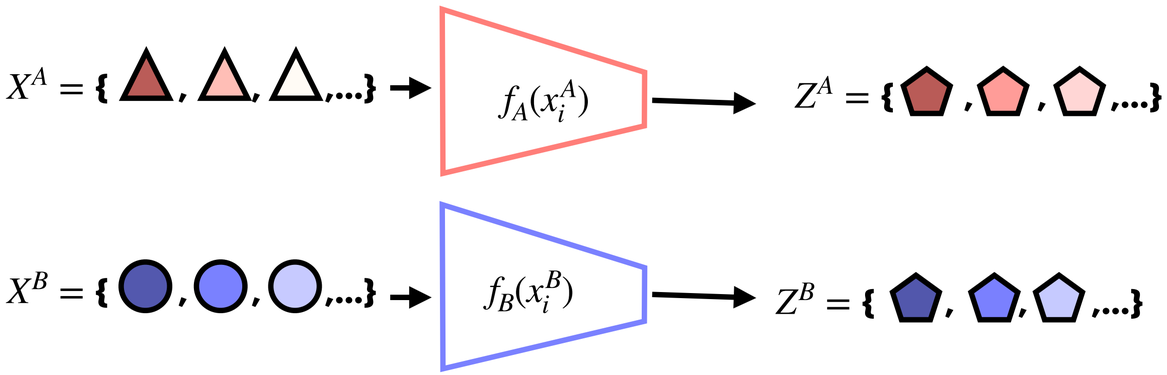
Data spaces Encoders Representation space

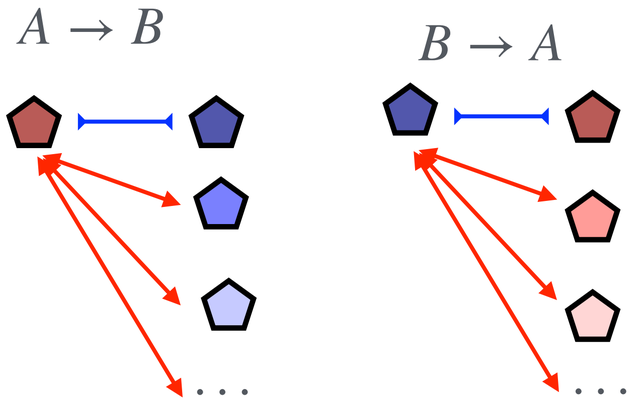


we define for each element of A as:

$$\mathscr{L}_{A\to B}^{(i)} = -\log \frac{\exp\left(\mathrm{sim}(z_i^A, z_i^B)/\tau\right)}{\exp\left(\mathrm{sim}(z_i^A, z_i^B)/\tau\right) + \sum_{j\neq i} \exp\left(\mathrm{sim}(z_i^A, z_j^B)/\tau\right)}.$$

Part2, establish correspondence: Train in both directions





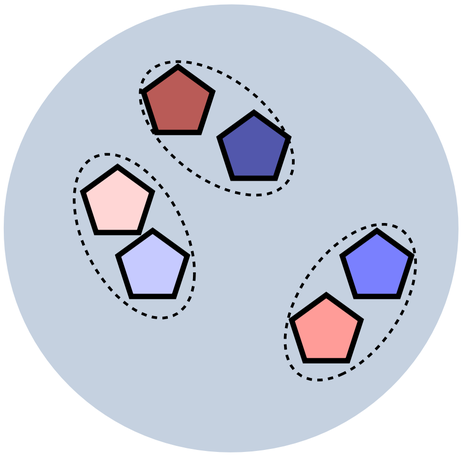
We train for each element and in both directions. This give "**InfoNCE**" Loss:

$$\mathcal{L}_{A\to B} = \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_{A\to B}^{(i)}$$

$$\mathcal{L}_{B\to A} = \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_{B\to A}^{(i)}$$

$$\mathcal{L}_{\text{align}} = \frac{1}{2}\left(\mathcal{L}_{A\to B} + \mathcal{L}_{B\to A}\right)$$

Training f_A and f_B



Recipe Card: **Paired Contrastive Alignment**

Type of Learning: **paired self-supervised**
Data type: **Paired multimodal data**

Data

1- Load libraries: Basics + PyTorch

2-Load paired data : $\mathscr{D}_{\text{paired}} = \left\{ \left(x_i^A, x_i^B\right) \right\}_{i=1}^{N}$

-Each pair means: $x_i^A \longleftrightarrow x_i^B$

3-Build mini-batches of paired observations

$$\mathscr{B} = \left\{ \left(x_i^A, x_i^B\right) \right\}_{i=1}^{m}$$

4- Within a batch, the matched pairs are the diagonal pairs:

$$(x_i^A, x_i^B), \qquad i = 1,\ldots,m\,.$$

5-Turn each pair into a retrieval question:

-For each x_i^A we compare with: $x_1^B, x_2^B, \ldots, x_m^B$.

-For each x_i^B we compare with: $x_1^A, x_2^A, \ldots, x_m^A$.

NN model

6- Use one encoder per modality

$$z_i^A = f_A(x_i^A)$$
$$z_i^B = f_B(x_i^B)$$

-Note: the architecture of f depends on each data.

- Both representations now live in: $z_i^A, z_i^B \in \mathbb{R}^k$

7- Compute cross-modal similarity:

$$S_{i,j} = (z_i^A)^\top z_j^B$$

Training

8-Within this mini-batch:

8.1-Retrieve modality B from modality A: $\mathscr{L}_{A\to B} = -\frac{1}{m}\sum_{i=1}^{m} \log \frac{\exp(S_{ii}/\tau)}{\sum_{j=1}^{m}\exp(S_{ij}/\tau)}\,.$

8.2-Retrieve modality A from modality B: $\mathscr{L}_{B\to A} = -\frac{1}{m}\sum_{i=1}^{m} \log \frac{\exp(S_{ii}/\tau)}{\sum_{j=1}^{m}\exp(S_{ji}/\tau)}\,.$

8.3-Calculate total loss: in both directions: $\mathscr{L}_{\text{align}} = \frac{1}{2}\left(\mathscr{L}_{A\to B} + \mathscr{L}_{B\to A}\right).$

9- Update the encoders:

$$\mathscr{L}_{\text{align}} \rightarrow \text{backpropagate} \rightarrow \text{update}\, f_A, f_B\,.$$

10- Repeat for every mini-batch and epoch.

11 - Evaluate on held-out paired observations:

11.1 For a held-out x_i^A, retrieve the predicted pair: $\hat{j} = \text{argmax}_j\, S_{ij}$.

11.2 The retrieval is correct when: $\hat{j} = i$.

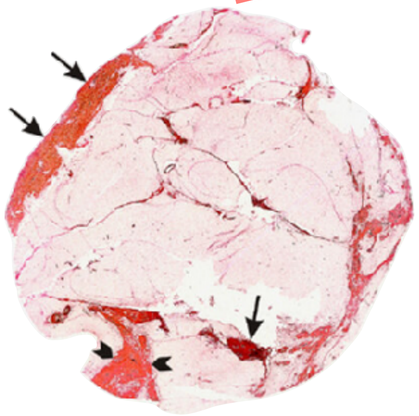
11.3 repeats for held-out x_j^B....

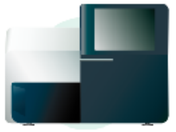
11.4 Report retrieval accuracy via @Recall curves

Toy example: Breast cancer-Visium

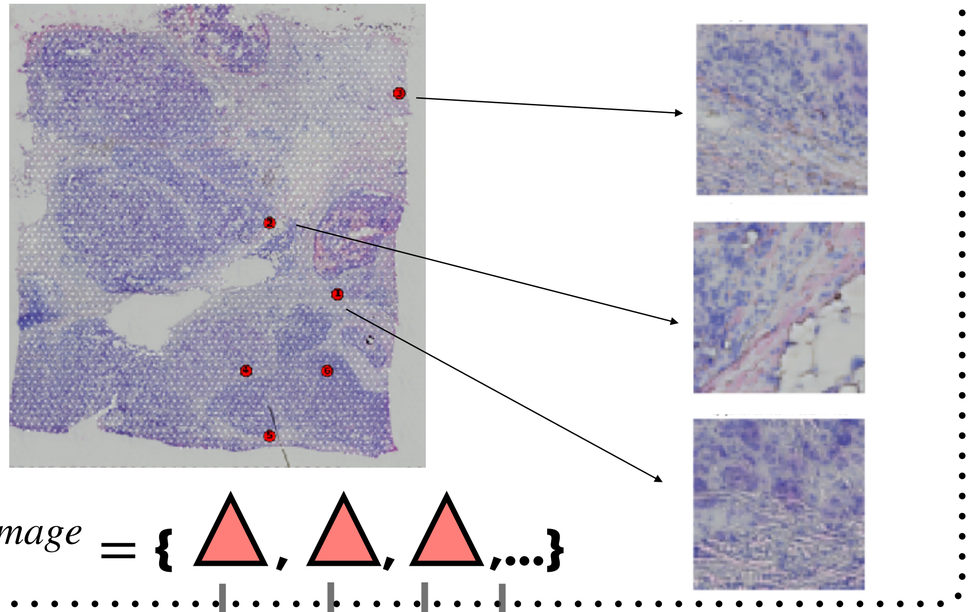


histology H&E





Spatial RNA-seq (Visium)



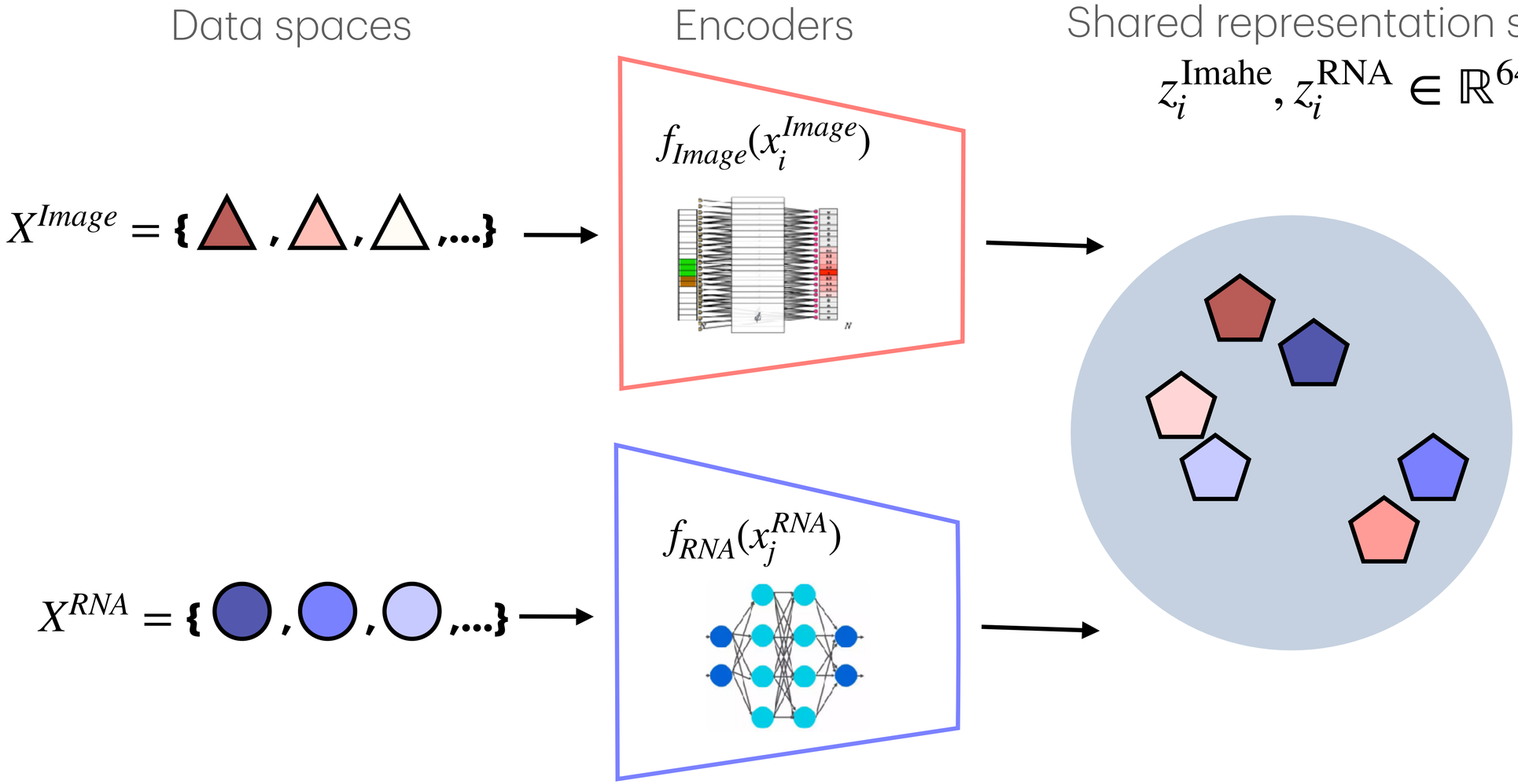
Toy example: Breast cancer-Visium

Human breast-cancer

Visium section H&E tissue image

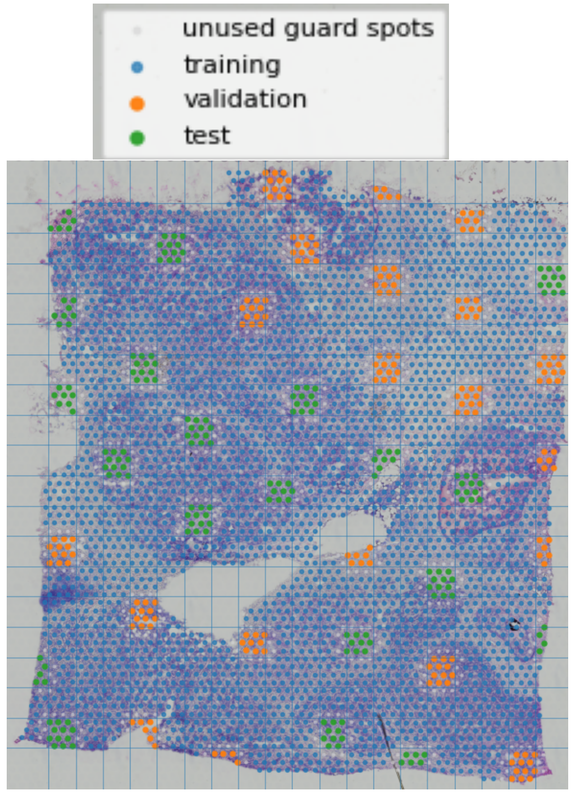
Spatial RNA profile at each Visium spot

N=3,798 paired spots



Toy example: Training

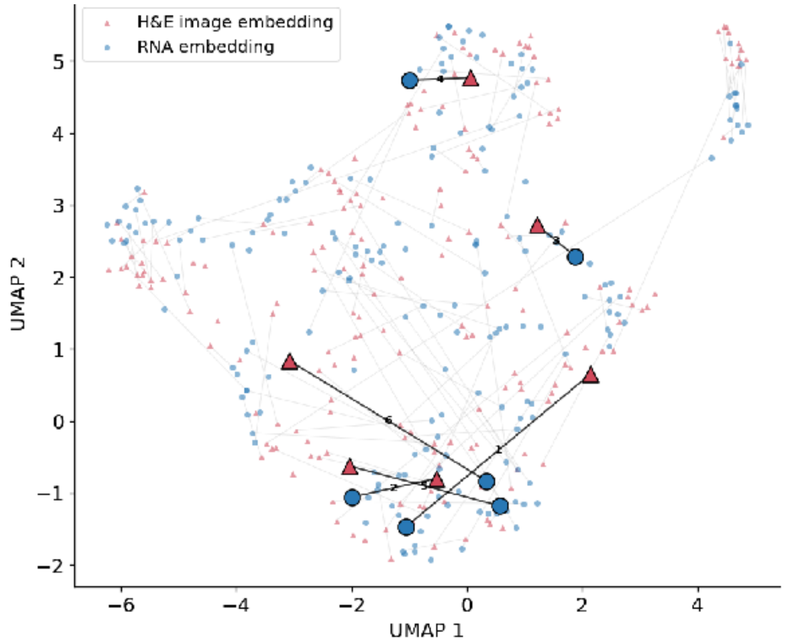
Spatial split



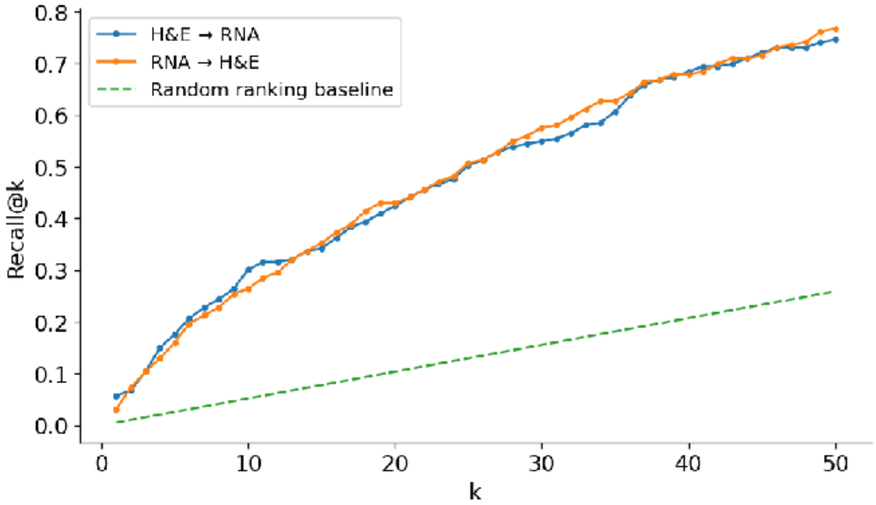
Training f_{Image} and f_{RNA} via

$$\mathscr{L}_{\text{align}} = \frac{1}{2}\left(\mathscr{L}_{A\to B} + \mathscr{L}_{B\to A}\right)$$

Learned shared space (held outs)

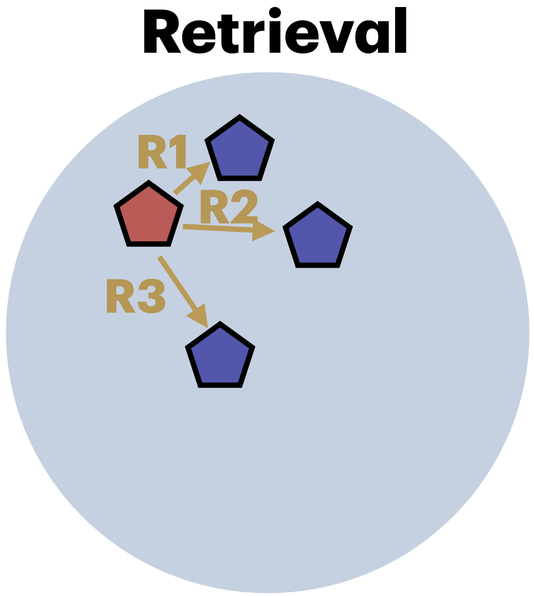


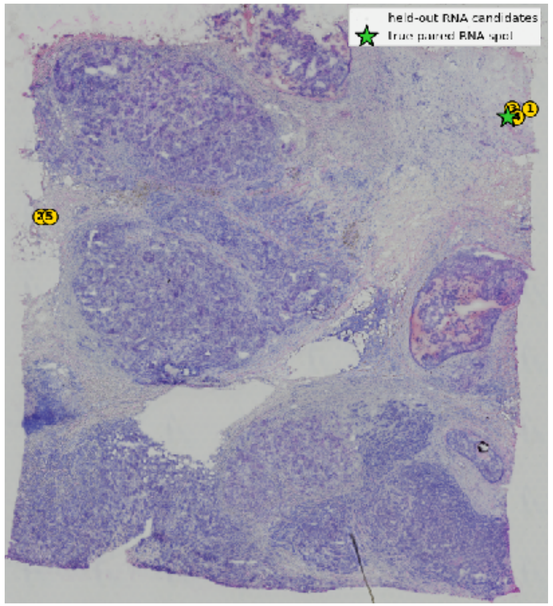
Given one modality, rank all held-out candidates in the other modality.

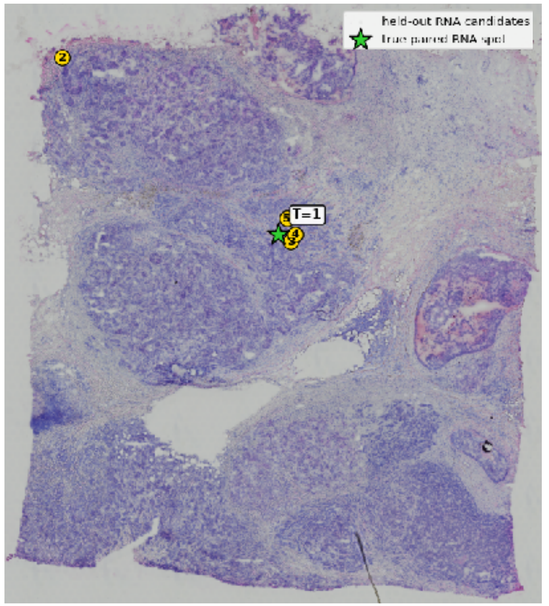


Toy example: Information retrieval examples

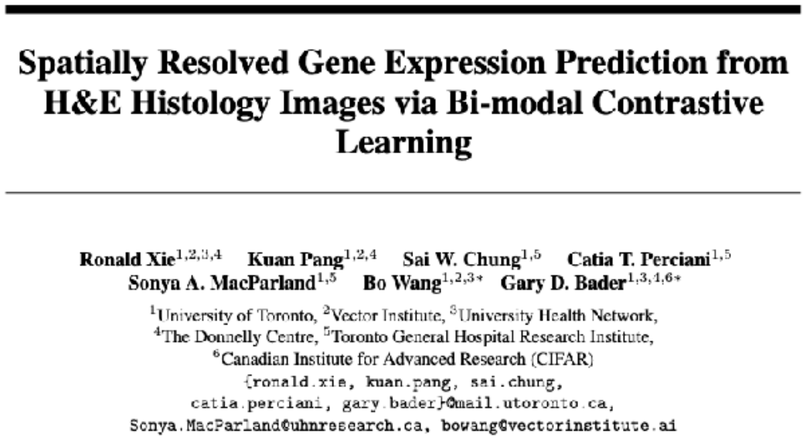
For each held-out image patch, we ask the model to rank held-out RNA profile.
The **green star** marks the RNA profile that truly belongs to that image location.
Yellow circles show the model's top-ranked RNA candidates: R1 is its best prediction, then R2, R3, and so on.







The model sometimes retrieves the exact paired RNA spot; when it misses, its leading candidates can still concentrate in the correct local tissue neighbourhood.



Spatially Resolved Gene Expression Prediction from H&E Histology Images via Bi-modal Contrastive Learning

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Read more on similar works:

Xie, Ronald, et al. "**Spatially resolved gene expression prediction from histology images via bi-modal contrastive learning**." Advances in Neural Information Processing Systems 36 (2023): 70626-70637.

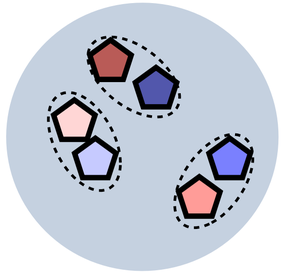
Min, Wenwen, et al. "Multimodal contrastive learning for spatial gene expression prediction using histology images." *Briefings in Bioinformatics* 25.6 (2024): bbae551.

Pang, Minxing, Kenong Su, and Mingyao Li. "Leveraging information in spatial transcriptomics to predict super-resolution gene expression from histology images in tumors." BioRxiv (2021): 2021-11.

What contrastive alignment actually learns? Caveats?

- Alignment gave us a task-agnostic shared space.
- it captured “some” of shared information between modalities,
- It can discard the modality-unique information U_A, U_B,
- is blind to synergy S_Y, and never optimized for any task Y.
- It does not capture interaction between modalities

Contrastive learning alignments learns
Information shared across paired modalities



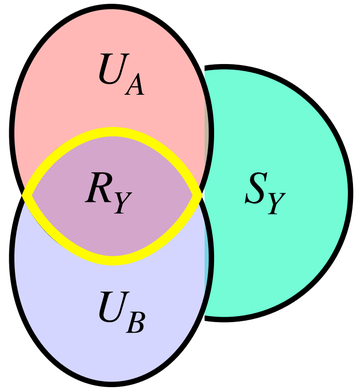
$$\mathcal{L}_{\text{align}} = \frac{1}{2}\left(\mathcal{L}_{A\to B} + \mathcal{L}_{B\to A}\right)$$

$$I(Z^A; Z^B)$$

$\not\Rightarrow$

Shared information between modalities does not automatically mean useful for the task Y.

$$R_Y$$



Radford et al., Learning Transferable Visual Models From Natural Language Supervision, ICML 2021

Liang et al., Factorized Contrastive Learning: Going Beyond Multi-view Redundancy, NeurIPS 2023